The textbook inequality is green. On a tissue of length π no admissible mode is in the band; at 2π one is.

Declared model: J = [[1, -2], [2, -3]], diffusivities D = (1, 10), 1-D Neumann boundaries; det(J − qD) = (5q-1)(2q-1), so the continuum band is 1/5 < q < 1/2. Admissible modes are q = (nπ/L)².

LENGTH L / Π	1/2	3/4	1	3/2	2	5/2	3	4	5	6
linear verdict	stable	stable	stable	grows	grows	stable	grows	grows	grows	grows

PROVED **COMPILED INSTANCE**

Lean 4.35.0: L = π admits no unstable mode; L = 2π admits one (n = 1, det = -1/8). Build exit 0; 3 mutants rejected. Only these two lengths are compiled.

EXACT ARITHMETIC

The whole strip is exact rational arithmetic. Mode n grows iff $2n^2 < (L/\pi)^2 < 5n^2$.

OBSERVED

None. No wet-lab experiment; simulations only.

Classical mathematics, a linear-mode statement for a declared model. It is not a claim that a pattern forms, and not that Turing was wrong. What is new is only that a program checks it, with a witness, before a law is applied.

A request the plain formula accepts is refused with a witness, repaired, and re-checked by an independent solver.

REQUEST

Target wavelength, equal diffusivities (Dv/Du = 1). 16 requests.

→

EXACT REFUSED

No admissible unstable mode (turing-ddi). 16 of 16.

→

EXACT REPAIR

Raise Dv/Du to 9 (most requests); the exact check licenses it.

→

SIMULATED SOLVER

Independent solver: pattern on 11 of 16 requests; mean success 0.708 (fraction of 3 noise seeds, averaged over requests).

The plain formula's design without repair succeeded on 0 of 16, by construction: equal diffusivities admit no instability for any kinetics.

Limitation first. The algorithm is trivial. The contribution is the certificate, the refusal and the repair.

CAVEATS, PRINTED

One kinetics, 1-D Neumann, 3 noise seeds, no confidence interval; a model-only check, not an experiment. Near-onset repaired patterns are marginal (amplitude up to 0.18 against a 0.10 threshold). The other 5 requests did not succeed on all seeds, and repairs for 2 further refusals worked on 0. No domain-length repair was found. Across 80 nonlinear configurations the gate agreed with the solver on 80 and the plain formula on 79; linear theory named the dominant wavelength in 43/61 patterned cases.

Ledger unledgered (NCA worker artifacts) · artifacts/rethink_nca/repair.json · scopedesk/oracles/results/linear_vs_nonlinear.json · planning/rethink/nca.md · **Review** single-reader, not Codex-reviewed: NCA worker artifacts and a critic audit of the checker code · **Sweep** preregistered, simulated, 2,520 cells of two textbook kinetics: 30 of 1,080 linearly stable cells (all Brusselator, large perturbation, just below onset) still patterned; 0 of 1,440 unstable cells failed to pattern; dominant wavelength differed from the fastest linear mode in 461 of 1,440; 59 unresolved cells are in every denominator. One author, a stated prior, not a claim about tissues; prediction P3 failed as stated; single-reader, not independently reviewed · scopedesk/oracles/results/nonlinear_sweep/REPORT.md · artifacts @ 1b1dc0b8

Two compiled nodes, and a test set whose answers come from stated methods; the checker is scored against it.

PROVED **PROOF DEPENDENCIES**

`price · ATLAS has it; ours duplicates` →

`selection_only` → `hamilton_regression`

Conditional Hamilton regression theorem. Lean exit 0, standard axioms only, 4 witness controls, 2 changed-sign mutants rejected. A critic's code search found no Hamilton form in ATLAS; not exhaustive. Second-reader review: **PENDING**

TEST SET

Version 1.2: **127** requests, **8** families, key hash 28645a12…. **Frozen.** Codex's metadata and wording check passes (14 of 14); it does not audit the answers. Answers: exact rational arithmetic for Turing, Hamilton and Hawk–Dove items; numerical or closed-form for Kimura, Greenhill and Fisher–KPP; all under declared conventions. Audit trail: a blinded payload-only audit agreed on 132 of 135 items of an earlier candidate, by a model of the same family as the key's author.

CHECKER AGAINST THE KEY

Turing, 2-D Turing and wavelength items.

METHOD	AGREES
Gate with the record	**50 / 50**
Plain textbook formula	29 / 50
Continuum condition only	34 / 50
Record hash altered: refuses	50 / 50

Printed caveat: the set is trap-enriched and shares its convention document with the checker. This is not external validation. The last row shows provenance dependence only.

The ledger: claims we held that did not survive, each with a witness and the kind of failure kept separate.

CLAIM WE HELD	KIND	WITNESS
Turing needs a diffusion ratio of at least 10	**counterexample**	a Lean-checked linear model with diffusivity ratio 1.1 has a growing mode (determinant -21/25000)
Pinned-end liquid bridge unstable beyond L > πr (our benchmark key)	**convention defect**	the n=1 mode violates the fixed-volume constraint; caught before any agent ran
Thompson's scaling refuted by a pooled regression	**mis-operationalised**	a pooled OLS tests mean allometry, not the buckling limit
Benchmark ids were label-neutral	**convention defect**	ids encoded generator order: a tree on id suffixes alone scored 58/96 against 43/96 for the majority answer
Our Price node as a novel formalization	**prior art**	atlas-lean already proves Price and Fisher; ours was a duplicate

21 ledger rows: 16 were caught by an independent reviewer or validator, 2 by the author with an independent hold, 3 only by the author's own measurement or rule. There is no control arm: this compares independent audit with self-audit. It is not a measurement of collective intelligence. Kinds: prior art 7, counterexample 1, measured 4, mis-operationalised 3, convention defect 4, process defect 1, design defect 1.

Also in the ledger: our theorem advertised as stability whose premise, determinant non-negativity, also holds for unstable systems (R06); a stronger statement now compiles (F10). Row R09, counted as measured, is a mismatched run (nonlinear prompts scored against linear labels), not a measured null. One team, one day: not a rate. No median is shown.

What the collective measured: no advantage shown in two small comparisons, and the proof pilot sat at a ceiling.

SIMULATED ENZYME REPLAY

159,129 measured variants, 1,000 extra assays.



No collective advantage shown. One repeated panel, about 24 distinct problems; the lazy search's gain sits mostly in one repeated pair and its assay-count interval includes zero.

HEADROOM CHECK ON THE FROZEN TEST SET

Unguarded models, no code tool: Claude Sonnet 5.5 **121/127** (95.3%), Gemini 3.8 flash **110/127** (86.6%). Checker-tool arm not run. Effective n is 8 families. This measures judgement, not derivation, and is a coarse consistency check scored by the key's author; not a model ranking. Gemini's misses sit in wavelength (7 of 14) and Greenhill (6 of 13). It mostly shows the test set is not hard for a frontier model without tools; nothing here says agents need the gate.

COMPILED PROOF-SEARCH PILOT, 19 FROZEN INTEGER-ARITHMETIC LEMMAS

Same model, 3 replicates, 30 calls per run: every arm certified 19/19 in every run, a ceiling. Library reuse is real but saved nothing: median calls none 21, private 22, sequential 20, shared 21; median tokens 72k, 80k, 78k, 80k. Deterministic tactics certified 4/19. All 228 recorded proofs re-certified after the checker was repaired, a repair made after the outcomes. Call-matched, not compute-matched; arithmetic surrogates, not biology.

Reusable now: a checker that returns a witness. Outside use: none observed yet.

NOW

The checker, repair enumerator, test set and ledger are in this repository. A fresh-clone acceptance run passed (exit 0, 56.6 s) at an earlier commit; a receipt for the final commit is PENDING

NEXT, A PROPOSAL

A pilot on a modelling team's recent reuse decisions. Delphi Frontier, a proposal, nothing deployed: one page per open problem showing what we know, what's next and who is working on it. Our own seed questions failed our own validator: 0 of 7 bounded (ledger R21, single-reader, not re-run by the critic). Misprediction map over a large ensemble: PENDING

CREDIT, AND WHAT WE DO NOT CLAIM

ATLAS (facebookresearch/atlas-lean), Prove2Me (arXiv 2608.28433), Rashid et al. 2017, lean-bio, crnt-lean, Ripple (arXiv 2607.13531). No first library, no universal threshold, and no claim that a named law is wrong.

SCORECARD

21 records: 6 linked to scope-check benchmark families, 1 related formal lemma (target unestablished), 2 empirical/framework, 12 without a corresponding check. Single-reader statuses, revised after a Codex review of version 0; the revision was not re-reviewed. The 12 are a backlog, not open problems, and the 6 are not 6 false laws.

Make the assumptions behind a biological-model decision executable and auditable.

Ledger A01 A02 P01 P07 P08 · scopedesk/scorecard/ (critic, v0) · scopedesk/oracles/results/accept_receipt_fresh_clone.json · planning/rethink/PRIOR-ART-VERDICT.md · **Infrastructure used, receipts only** Nebius H100s with NVIDIA BioNeMo ESM2: 5 sizes, 10 jobs, 9,475 recorded job-seconds, no clear benefit (B03) · Nebius hosted OpenFold2: 2 structure calls, predicted, low confidence · OpenResearch: local CPU analysis logs (enzyme data; nonlinear sweep) and full-text retrieval for citation verification · Gemini: 4 bounded review calls and the headroom run · receipts in artifacts/scale/ artifacts/services/ artifacts/gpu/orx/ scopedesk/ledger/orx_literature_receipt.json · **Review** single-reader, not Codex-reviewed: scorecard statuses (revised after a Codex review of the first draft; the revision was not re-reviewed) · artifacts @ 1b1dc0b8